

**Time:** 5 minutes**Outcome:** Analyze Profit-Maximizing Output & the $MR = MC$ Rule.**Difficulty:** ●●○ Intermediate

Profit-Maximizing Output & the $MR = MC$ Rule



THE CORE IDEA

A firm maximizes profit or minimizes loss by producing the quantity where rising marginal cost equals marginal revenue. The local profit maximum occurs where MC rises through MR . Marginal benefit to the firm exceeds marginal cost. Market price becomes the firm's MR , linking market equilibrium to the firm's marginal output choice.



HOW TO RECOGNIZE IT

- Use the firm's marginal-revenue curve and the rising portion of marginal cost.
- Choose the output where marginal cost equals marginal revenue, provided the firm should operate.
- If marginal revenue exceeds marginal cost, another unit raises profit; if marginal cost exceeds marginal revenue, it lowers profit.
- Do not choose output by minimizing average total cost unless that output also satisfies the marginal rule.

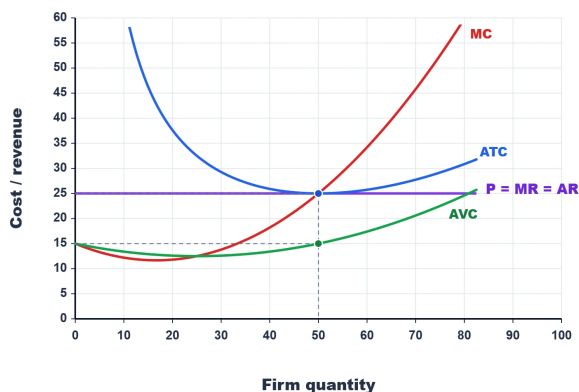


WATCH OUT

A common mistake is to misapply competitive firm rule. Market price becomes the firm's MR , linking market equilibrium to the firm's marginal output choice.



WORKED EXAMPLE: USING THE $MR = MC$ RULE



The firm's market price and marginal revenue are \$25. The rising marginal-cost curve reaches \$25 at 50 units, so the last unit adds exactly as much revenue as cost. Producing fewer units leaves profitable output unmade; producing more makes MC exceed MR . Profit-maximizing output is 50.



CHECK YOURSELF

MR is \$25 and the next unit's MC is \$19. What should the firm do?

**READY?**

Return to the game and master this concept.